

CSE 598 GBL Benchmark Challenge

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Abstract

This experiment studies protein generation across lengths from 90 to 128 amino acids. We adopt a de novo protein backbone generation pipeline where we first use RFDiffusion for backbone generation, then use ProteinMPNN for sequence design and ESMFold for structure prediction. We compare three RFDiffusion sampling configurations: default sampling, a fast reduced-timestep setting, and a very low-noise setting. The generated proteins are evaluated using designability, diversity, novelty, and secondary structure diversity. Overall, the results show a clear trade-off between designability and diversity: the very low-noise setting achieves the highest designability, while the fast setting produces the largest number of structural clusters.

1 Introduction of the Challenge

The goal of this protein design challenge is to generate 10 protein designs for every length from 90 to 128 amino acids. We aim to produce proteins that are structurally valid, sequence-designable, diverse, and novel relative to known proteins.

2 Methodology for Protein Generation

2.1 Overall Pipeline

We use a two-stage de novo protein design workflow [3] consisting of backbone generation, sequence design, structure prediction. First, we use RFDiffusion [3] to generate protein backbone structures. Then, ProteinMPNN [1] is used to design amino acid sequences for each generated backbone. Finally, we use ESMFold [2] to predict the folded structure of each designed sequence.

2.2 Backbone Generation with RFDiffusion

RFDiffusion is used to generate protein backbones for all lengths from 90 to 128 amino acids. For each length, we sample 10 backbones, resulting in approximately 390 generated structures per configuration. See Figure 1 for the illustration of an example.

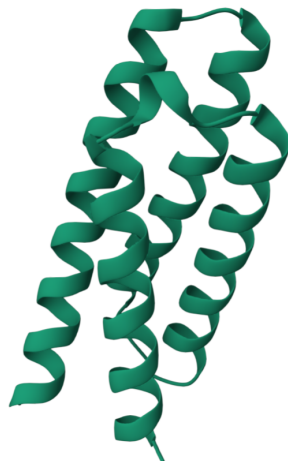


Figure 1: An illustration of a generated protein backbone using RFdiffusion

We compare three RFdiffusion configurations. The first is the default RFdiffusion setting. The second is a low-noise setting, where we use *denoiser.noise_scale_ca=0.5* and *denoiser.noise_scale_frame=0.5*. This setting is expected to improve structural quality and designability, but may reduce diversity. The third is a fast setting, where we use *diffuser.T=20*. This setting reduces the number of diffusion steps and is expected to improve efficiency, but may lower structural quality.

2.3 Sequence Design with ProteinMPNN

For each RFdiffusion backbone, we use ProteinMPNN to design amino acid sequences, which is an inverse folding model that predicts sequences likely to be compatible with a given backbone. We generate three sequences for each backbone to increase the chance of finding at least one sequence that can fold back into the intended structure.

2.4 Structure Prediction with ESMFold

Finally, each designed sequence is folded using ESMFold. We compare the predicted structures with the original RFdiffusion backbones. This step tests whether the designed sequences can recover the intended generated structures.

3 Evaluation Setup

We evaluate each configuration from four perspectives: designability, diversity, novelty, and secondary structure diversity. Metrics other than designability are computed on designable samples..

Table 1: Comparison of protein generation methods under designability, diversity, structural diversity, and novelty metrics.

Method	Designability		Diversity	Structural Diversity			Novelty
	%Designable ($\uparrow$)	pLDDT ($\uparrow$)	MaxCluster ($\uparrow$)	%Coil	%Helix	%Strand	TM-score ($\downarrow$)
default	94.9	88.4	119	15.3	75.5	9.2	73.8
fast	87.4	84.7	149	17.6	69.9	12.5	72.1
very low noise	99.5	92.0	29	11.2	87.7	1.1	80.6

3.1 Designability

Designability measures whether a generated backbone can be used to design amino acid sequences. For each backbone, we fold the three ProteinMPNN-designed sequences with ESMFold and compute the RMSD between each predicted structure and the original backbone. We define scRMSD as the minimum RMSD within the three predicted structures. We define the designability ratio as the fraction of generated samples whose its scRMSD is below 2.0.

We also measure ESMFold pLDDT that tells the confidence of the predicted fold.

3.2 Diversity

Diversity measures how many distinct the generated structural groups are. We use Foldseek to cluster designable structures with *-alignment-type 1*, *-tmscore-threshold 0.5*, and *-tmscore-threshold-mode 0*. The number of Foldseek clusters is used as the diversity metric. More clusters indicate higher structural diversity.

3.3 Novelty

Novelty measures how unique the generated proteins are from known protein structures. We use Foldseek to search each designable sample against the PDB database. For each sample, we record the highest TM-score to any PDB chain, denoted as pdbTM. Lower pdbTM indicates higher novelty, and vice versa.

3.4 Secondary structure diversity

Secondary structure diversity describes the structural content of the designable samples. For each generated protein, we compute the percentages of helix, beta strand, and coil residues.

4 Results

Table 1 compares the three RFdiffusion configurations on our four perspectives. Overall, these metrics show different trade-offs between designability, diversity, and novelty.

The fast setting has the lowest designability, with 87.4% designable samples and a pLDDT of 84.7, likely because fewer diffusion steps reduce the quality of the backbones. However,

it achieves the highest diversity, with 149 clusters, and the lowest PDB TM-score of 72.1, suggesting better novelty.

The very low-noise setting achieves the strongest designability, with 99.5% designable samples and the highest pLDDT of 92.0. This suggests that reducing the denoising noise improves structural self-consistency. However, it only produces 29 MaxCluster clusters, far fewer than the other methods, indicating reduced structural diversity.

Secondary structure diversity also reflects these differences. The very low-noise setting is highly helix-rich, with 87.7% helix and only 1.1% strand, while the fast setting has more coil and strand content. Again, it shows that fast setting demonstrates higher diversity than very low-noise setting.

4.1 Trends with Protein Length

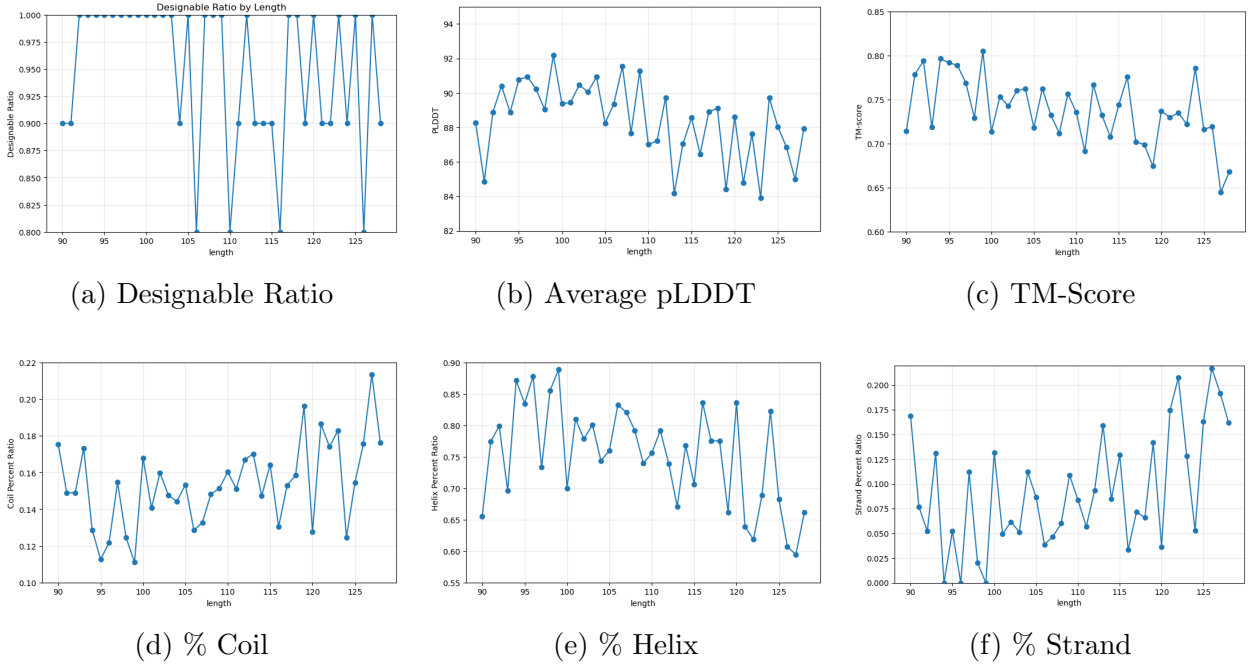


Figure 2: The changes of the protein quality as lengths differ.

Figure 1 shows that the default RFdiffusion setting performs well across all tested lengths. The designable ratio in Figure 1(a) and the average pLDDT in Figure 1(b) remains high for most lengths, usually close to 0.9 or 1.0, which indicates stable generation quality across the range from 90 to 128 residues.

In Figure 1(c), the TM-score tends to decrease slightly as length increases. Since lower TM-score indicates greater novelty in our setup, this suggests that longer proteins are relatively more novel than shorter ones.

The secondary structure plots show a gradual shift with length. The helix fraction in Figure 1(e) tends to decrease for longer proteins, while coil and strand content in Figure 1(d) and Figure 1(f) tend to increase. This indicates that shorter proteins are more helix-dominated, whereas longer proteins have more mixed secondary structure.

References

- [1] Justas Dauparas, Ivan Anishchenko, Nathaniel Bennett, Hua Bai, Robert J Ragotte, Lukas F Milles, Basile IM Wicky, Alexis Courbet, Rob J de Haas, Neville Bethel, et al. Robust deep learning-based protein sequence design using proteinmpnn. *Science*, 378(6615):49–56, 2022.
- [2] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Sal Candido, et al. Language models of protein sequences at the scale of evolution enable accurate structure prediction. *BioRxiv*, 2022:500902, 2022.
- [3] Joseph L Watson, David Juergens, Nathaniel R Bennett, Brian L Trippe, Jason Yim, Helen E Eisenach, Woody Ahern, Andrew J Borst, Robert J Ragotte, Lukas F Milles, et al. De novo design of protein structure and function with rfdiffusion. *Nature*, 620(7976):1089–1100, 2023.